Introduction

If you're reading this, that likely means you're engaging with the Regional Monitoring Programs map that the GitHub repository you found this document in is built around. The following text outlines some of the ideas the original creator of the map had for where this project could go next, as well as thoughts on how to envision new applications of this general architecture.

Motivation

The original motivation for developing this map came out of a desire to better understand the data systems that the California State Water Board works with, build familiarity with retrieving and inspecting that data, and then produce a deliverable that shared insight about it. The scope of the original map was intentionally kept small. For instance, it only maps the station footprint of 6 regional monitoring programs in California, and it only attempts to answer *where in California do these programs measure and collect data?*

Through this project, the gaps in the data became apparent, and I've done my best to document them within the project itself. When the "Summary" button is clicked, it walks through the steps used to trace the original data sources from the respective data collection websites through to how they were configured for downloading. A couple of examples are worth calling out. The Russian River Regional Monitoring Program, for instance, did not yield usable data from CEDEN the way the other programs did, so I instead used the shapefile for the Russian River streams themselves to generate the H3 hexagon tiles. Similarly, the Klamath Basin had clean station data, but only as a separate Microsoft Excel workbook downloadable from their own website rather than through CEDEN. These examples illustrate the unpredictable nature of data management across programs, and point to a broader need to understand, address, and remedy these inconsistencies in order to move toward a best-practice data management culture.

Ideas for What Comes Next

There are a few directions future interns, or anyone replicating a similar project, could consider.

1. Bring in the actual water quality data. This map currently only records static station footprint data but the actual water quality data collected at those stations (biological, toxicological, chemical, etc.) have intentionally been screened out at the point of the cleaning scripts. However, it wouldn’t take much to modify the scripts to ingest the actual data collected from each station, but the difficult would lie in handling the varied data schemas each dataset presents and finding a way to develop a shared language for the data that extracts the essence of what each data entry is saying about water quality at that location.

The complexity here lies mainly in the temporal element: a single station produces many data points over time, and there isn't a single consistent schema or format that this data is presented in across programs. Cleaning and structuring that temporal water quality data and then displaying it on the map so that a viewer could click on a station and toggle a timeline to see the actual readings collected there would be the natural next step toward an informative water quality map. One thing to note here though is you'd be limited by the availability and recency of data within portals like CEDEN. If a dataset is only updated every two years, then any data pulled from it will, by nature, be up to two years old. This doesn't rule out working toward a live data map, though, especially if participating organizations or remote sensing devices can eventually upload data on a faster cadence. That's just not the current state of things.

2. Expand beyond the original 6 programs. Another direction is to expand this same process to significantly more programs and datasets, rather than just the original 6 regional monitoring programs. In other words, any dataset about California water quality that exists as a substantial aggregation could be mapped and visualized in this same H3 hexagon tile format, giving a much fuller sense of the footprint of all the datasets we have access to, say, everything within CEDEN. The general process is: raw data ingestion with proper documentation, a custom script to clean the data into our repo's map-ready format, and then running that script to produce the cleaned CSV that the map reads. This wouldn't be difficult to replicate across all programs and datasets within a given data portal, and could even extend to bringing together datasets from multiple data portals. This would be a valuable step both in visualizing program coverage across California and in identifying data gaps which involve both flagging datasets that are insufficient or poorly maintained and understanding which parts of California have outdated or insufficient access to water quality data.

Conclusion

All in all, there are broadly speaking two directions to expand this project. One is diving deeper into the water quality data at each existing location. The other is expanding the number of programs and datasets visually displayed. Both are valuable in their own right, and both move this project closer to being a genuinely useful tool for understanding water quality monitoring coverage, and its gaps, across California.